

Skills

Languages: Scala; Java; TypeScript; Python; C; Bash; Cypher; SQL; HTML; CSS; Dhall

Frameworks/Tools: Git; Angular; React; PyTorch; Playwright; SBT; Maven; Gradle; Conda

Products: GitHub; Neo4J; PostgreSQL; MongoDB; Redis; Auth0; SendGrid; Typst

DevOps & Cloud: Kubernetes; Docker; GitHub Actions; Argo CD; Google Cloud; AWS

Experience

Full-Stack Software Engineer | Garner Distributed Workflow | Full-time 05/2024 – 09/2025

- Performed **prod db migration** (>1 billion nodes) to OmgDB (Open-source Neo4J alternative) which I maintained. Included **k8s/cloud** infrastructure, benchmarking, cost analysis, and query optimization.
- Designed a logistical filtering DSL, implemented in-memory and **cypher** filtering, and strongly typed **graph-ql** schema. Wrote full conversion layer and performed migration from existing filter system.
- Wrote Scala import cronjobs for importing millions of rows of financial and logistical data into backend.

Machine Learning Researcher | Pardee Lab | Full/Part-time 05/2023 – Present

- Implemented a distributed computing platform on **Google Cloud** using **MongoDB** to run a PyGAD genetic algorithm with a **PyTorch** VAE + LLM fitness predictor, reducing the running time by a factor of ~20.
- Researched biology and machine learning, monitored and scaled team progress to submit two papers at **ICLR 2025** (openreview.net/forum?id=sfTXliu7Op, openreview.net/forum?id=KBMxaCSwpB)

Projects

Recipe Website | github.com/SpecialThing44/recipe-website 2025

- Personal project for keeping track of recipes. Social network design using Neo4J and strong typing to allow analytical queries to power user-tailored recipe recommendation and powerful filtering engine.
- Restful, Scala Play backend based on functional programming and ZIO with Angular frontend.

Interactive Research Platform | University of Toronto 2023

- Worked in an Agile team to make a **React/Java** web platform that facilitates the real-time interaction of humans and software agents in decision games using a microservice architecture for games.
- Created a modular API for website-game interaction, designed a **Python** framework for researchers to create their own games and reinforcement learning models, and stored the data in **MySQL**.

Java Tamagotchi Game | University of Toronto 2022

- Collaborated with a team to create a Tamagotchi-style videogame in **Java** with a **Swing** UI following **MVC**, and **SOLID** Principles, and a database for online user interaction and saved games.
- Handled input processing and passthrough (Controller), serialization, and most tests with Junit.

Predictive Model with Web UI | University of Toronto 2021

- Utilized **Python** with Pandas and NumPy to aggregate real-world data and create a **SKLearn**-based predictive model to assess the impact of COVID-19 restrictions on restaurants for a formal report.

Education

University of Toronto | Toronto Ontario | Year 4 05/2026 (Expected)

- Bachelor of Science, Computer Science Specialization. A&S Internship Program.
- Relevant Coursework: Operating Systems; Algorithms; Machine Learning; Software Engineering; Web Programming; Software Testing; Databases; Data Structures